

Basic Principles of Medicine 1
Module: Musculoskeletal System | DLA

Clinical Anatomy of the Ankle and Foot, Arterial Supply, Venous and Lymphatic Drainage

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Recommended Reading

- Drake, Vogl, Mitchell. Grays anatomy for students. Newest ed. Elsevier Churchill Livingstone.
- Online Textbook: Chapter 6
 - Regional Anatomy
 - Foot and all topics related to it
- Pay special attention to the green “In the Clinic” boxes for the clinical correlations for this topic



Objectives

SOM.MK.II.BPM1.1.MSK.1.ANAT.1325	Describe the movements that occur at the ankle joint
SOM.MK.II.BPM1.1.MSK.1.ANAT.1326	Describe the ankle joint its and ligamentous support
SOM.MK.II.BPM1.1.MSK.1.ANAT.1327	List the two ligaments most commonly damaged during an inversion sprain.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1328	Describe the medial tarsal tunnel and its contents.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1329	Describe the general arrangement between the tarsal bones and the metatarsals
SOM.MK.II.BPM1.1.MSK.1.ANAT.1330	Describe hallux valgus and hallux varus.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1331	Describe the movements that occur at the tarsometatarsal, metatarsophalangeal and interphalangeal joints of the big toe and of digits 2-5.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1332	Describe the general attachments of the muscles of sole of the foot, their actions and neurovascular supply.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1333	Describe the cutaneous innervation of the thigh, leg and foot.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1334	Predict the regional lymph nodes which will be most likely enlarged because of infection(s) in the lower limb.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1504	Describe the arterial supply, venous and lymphatic drainage of the lower limb



Overview

- Ankle joint
 - Structure
 - Ligaments
- Bones of the foot
- Arches of the foot
- Actions of the joints of the foot
- Medial tarsal tunnel
- Plantar surface of the foot
 - Muscles and innervation
- Cutaneous innervation of the foot
- Blood supply of the foot
- Venous drainage of the foot
- Summaries
 - Blood supply of lower limb
 - Venous drainage of lower limb
 - Lymphatic drainage of lower limb
 - Cutaneous innervation of lower limb
 - Dermatomes of lower limb
 - Myotomes of lower limb

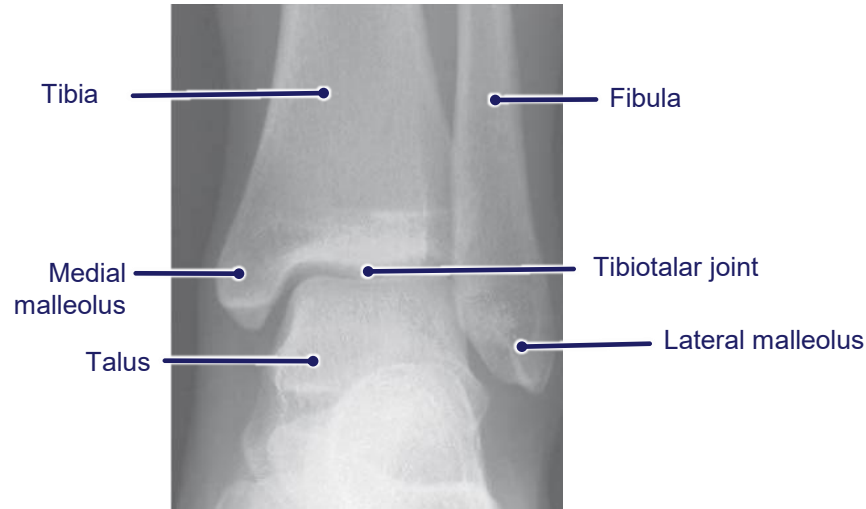
Actions of the Ankle joint

- Moving foot

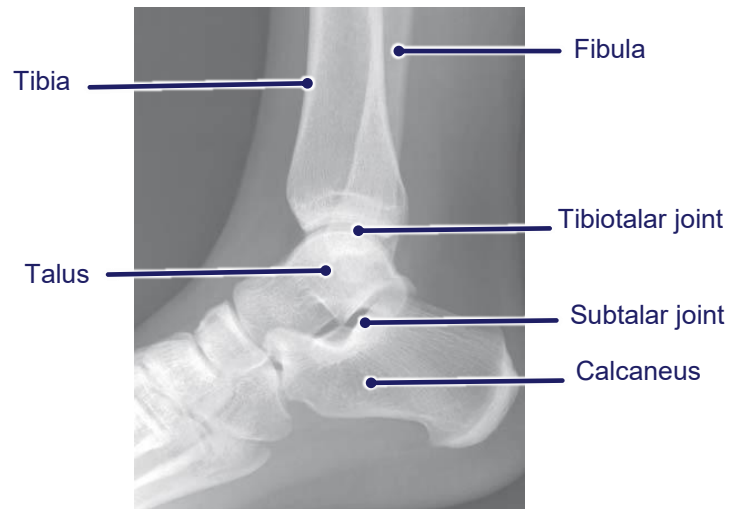
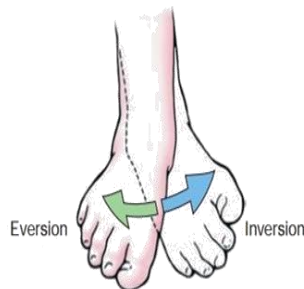


Structure of the Ankle Joint

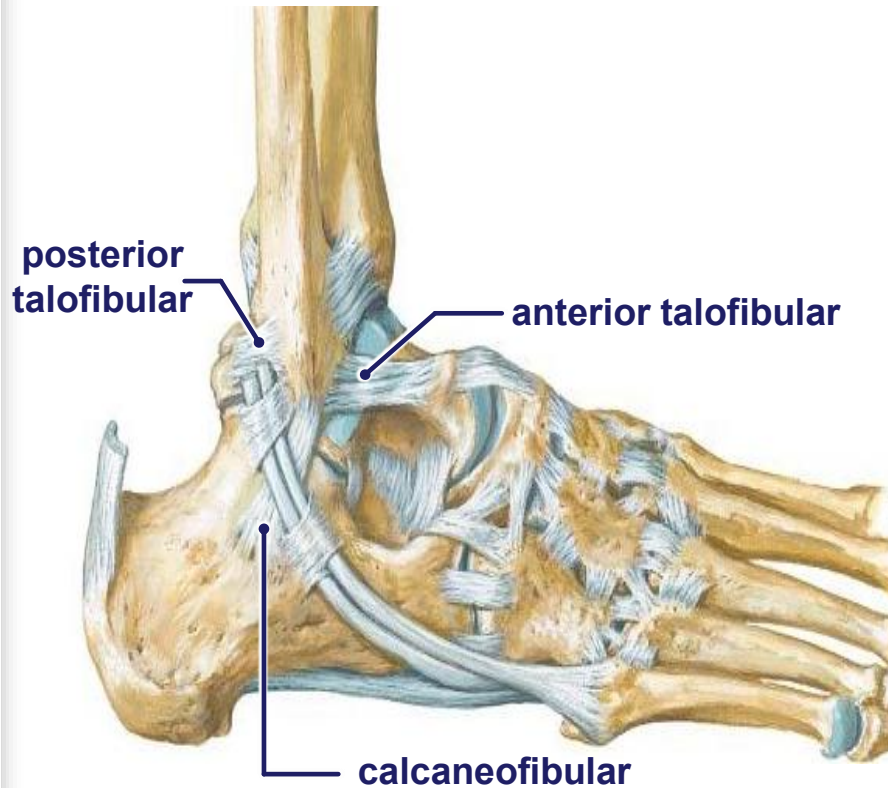
- Ankle joint is a synovial joint complex
- **Tibiotalar** joint allows hinge-like dorsiflexion and plantar flexion



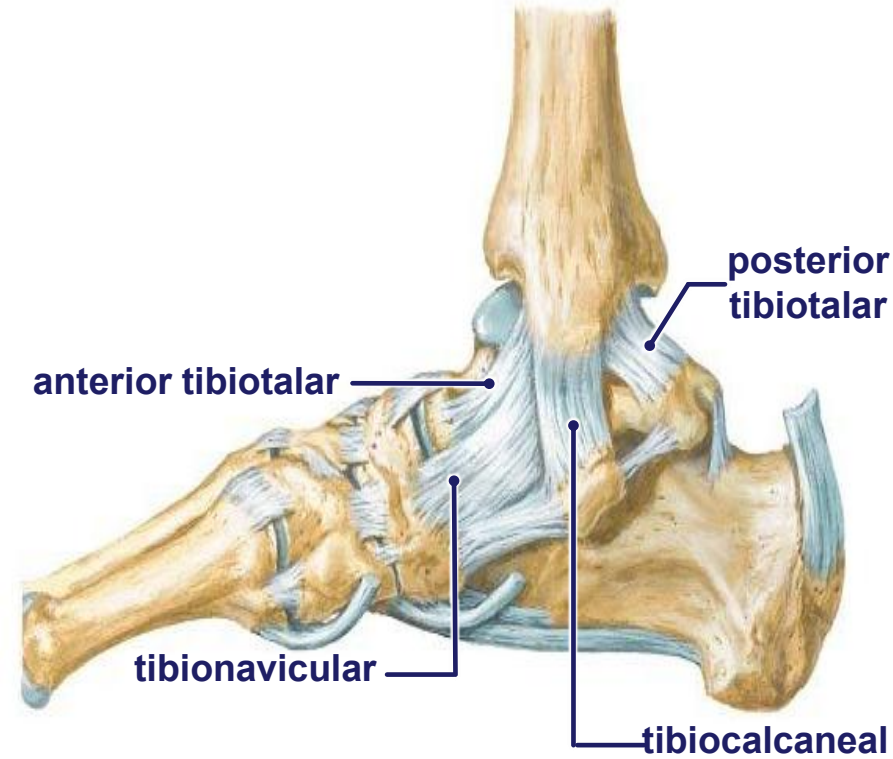
- **Subtalar** joint is a plane joint for inversion and eversion



Ligaments of the Ankle Joint

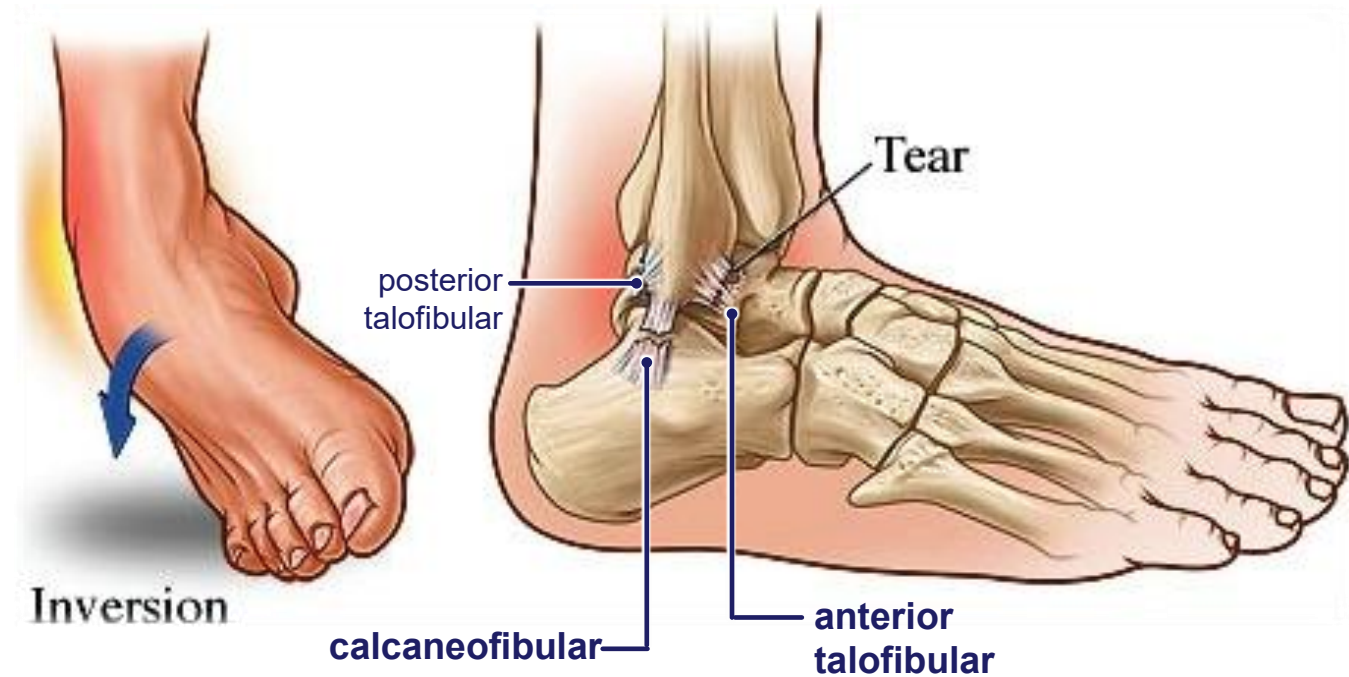


- Lateral ligament of the ankle is composed of 3 separate ligaments



- Medial (deltoid) ligament is large, strong and triangular in shape and composed of 4 separate ligaments

Inversion Injuries Stress the Lateral Ligaments

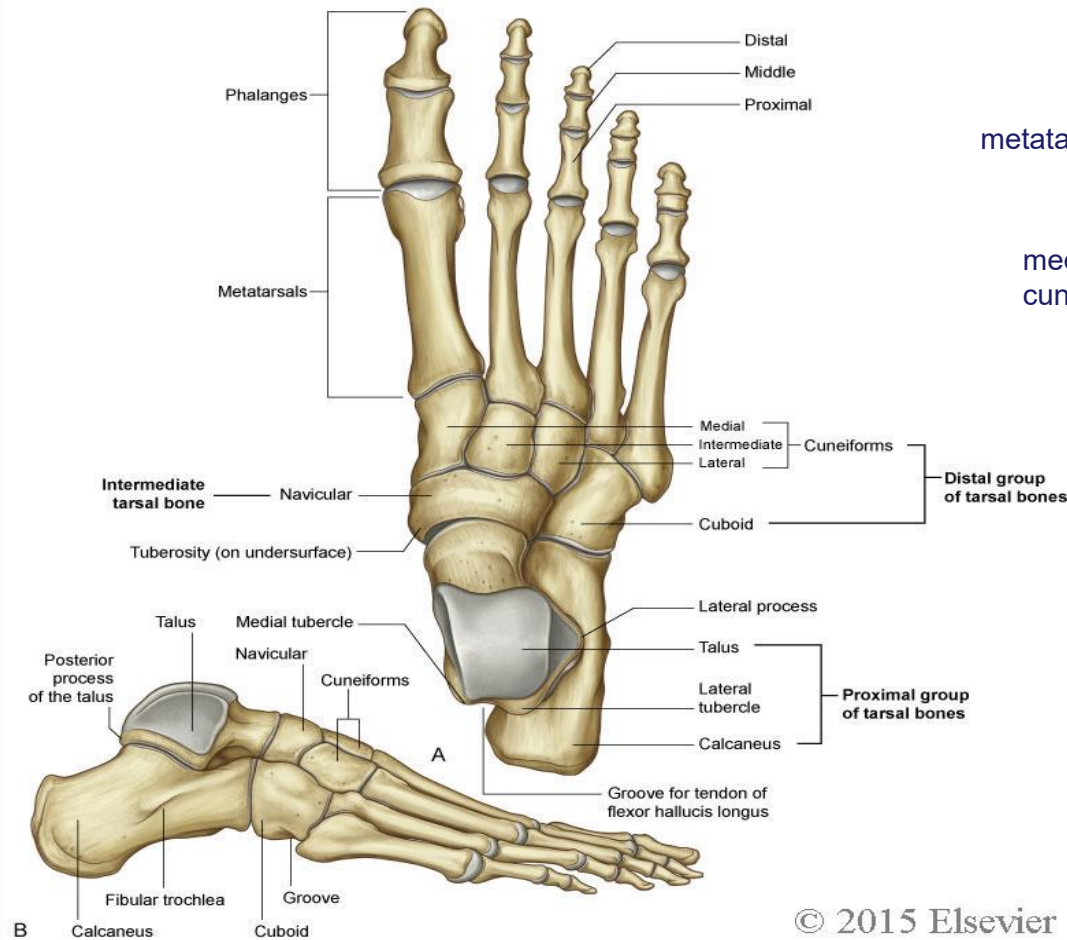
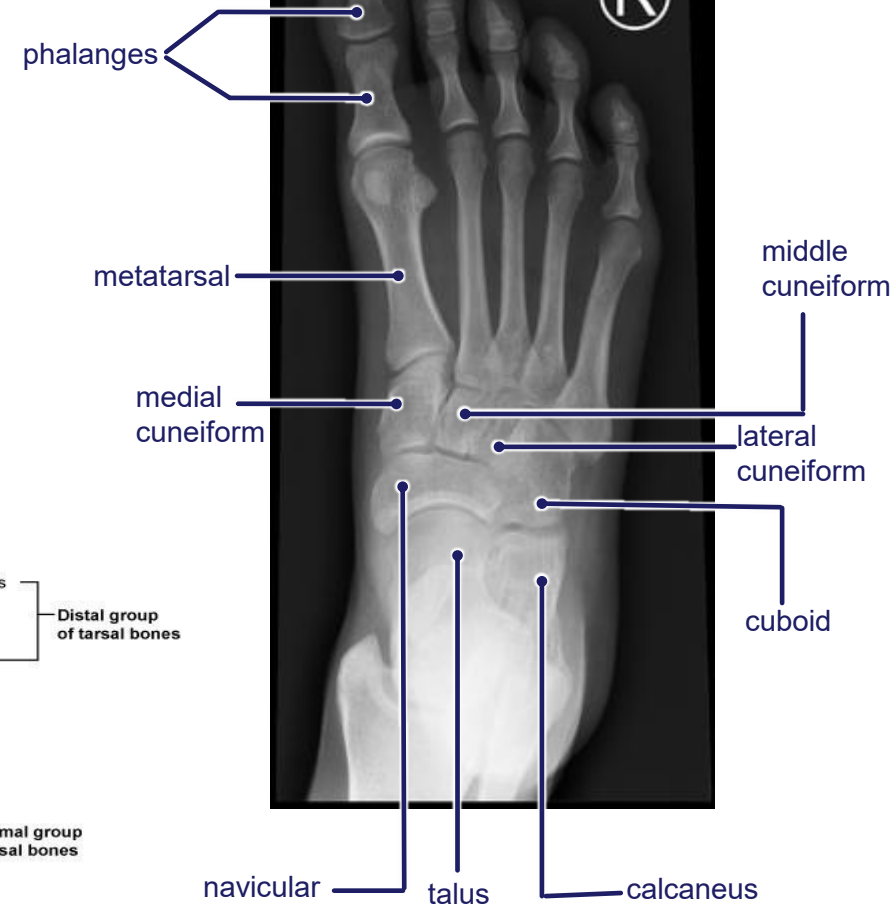
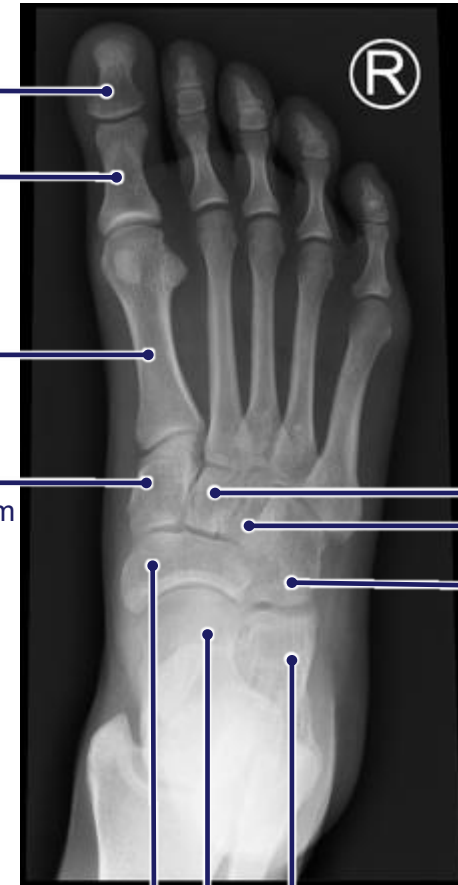


- Anterior talofibular and calcaneofibular most commonly torn in inversion stress to the ankle.

Bones of the Foot

- 7 Tarsal bones
- 5 metatarsal bones
- 14 Phalanges

https://radiopaedia.org/images/12509476?case_id=36688

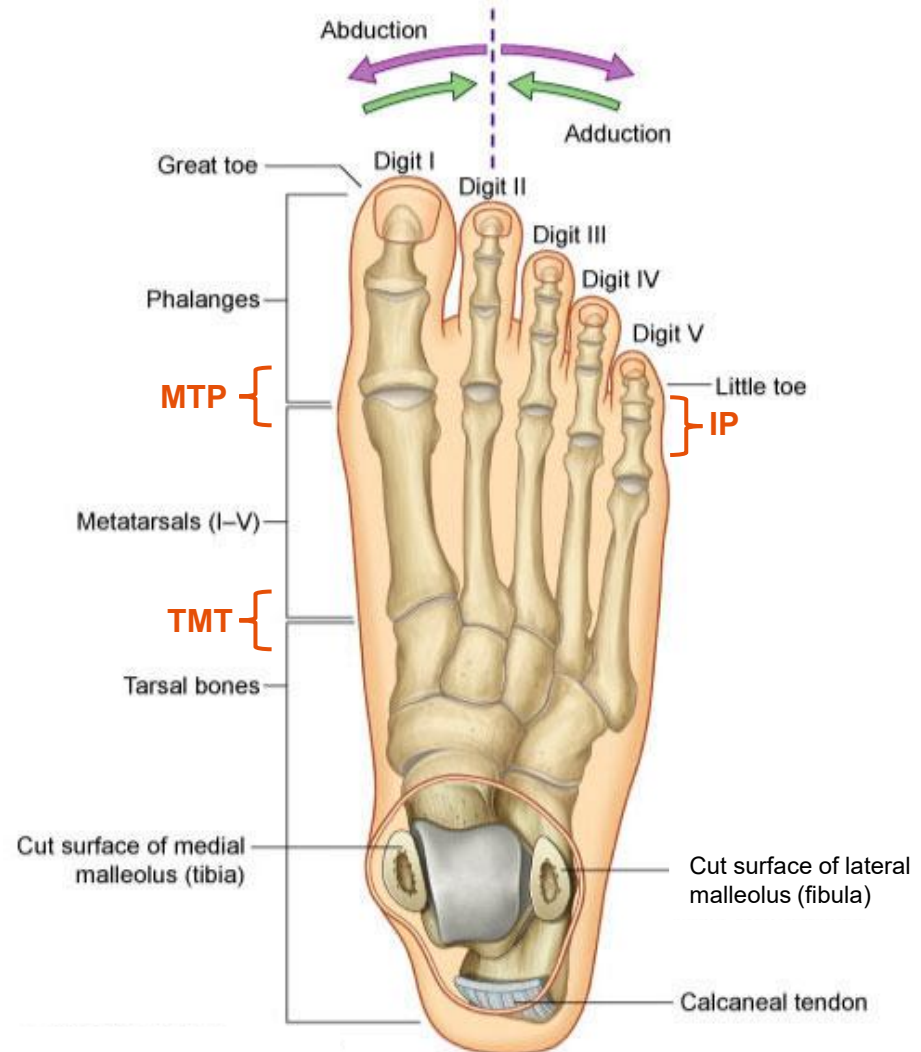


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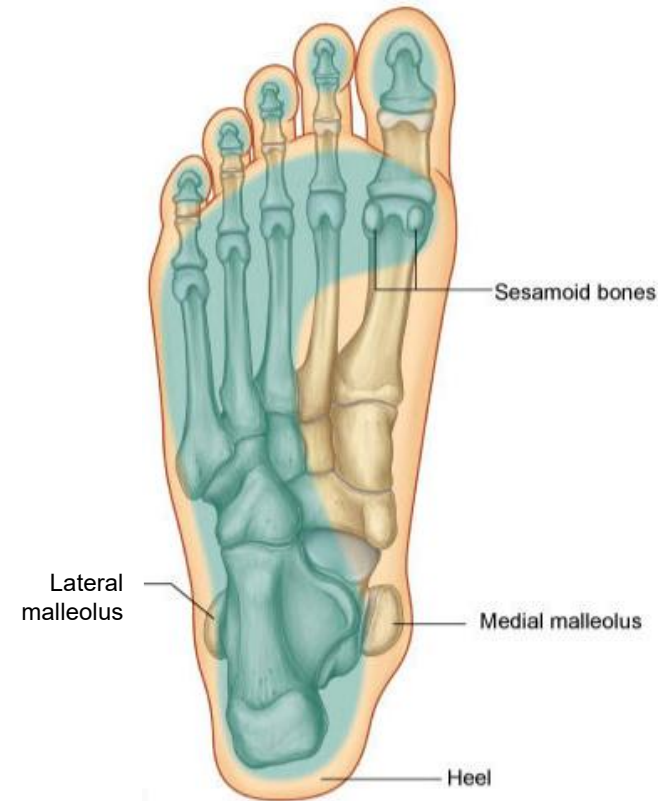
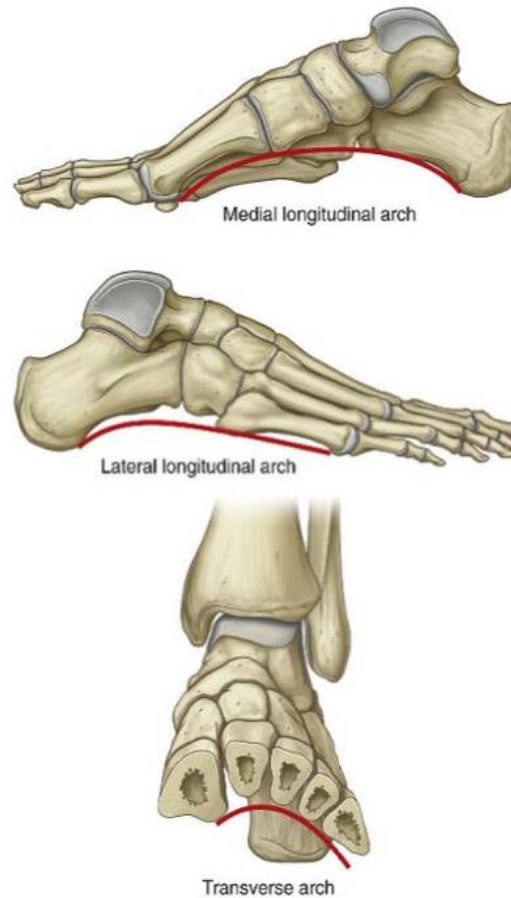
Actions of the Intrinsic Joints of the Foot

- Moving the toes
- **Tarsometatarsal (TMT) joints**
 - Fairly rigid, plane synovial joints with little movement
- **Metatarsophalangeal (MTP) joints**
 - Flexion
 - Extension
 - Abduction
 - Adduction
- **Interphalangeal (IP) joints**
 - Flexion
 - Extension

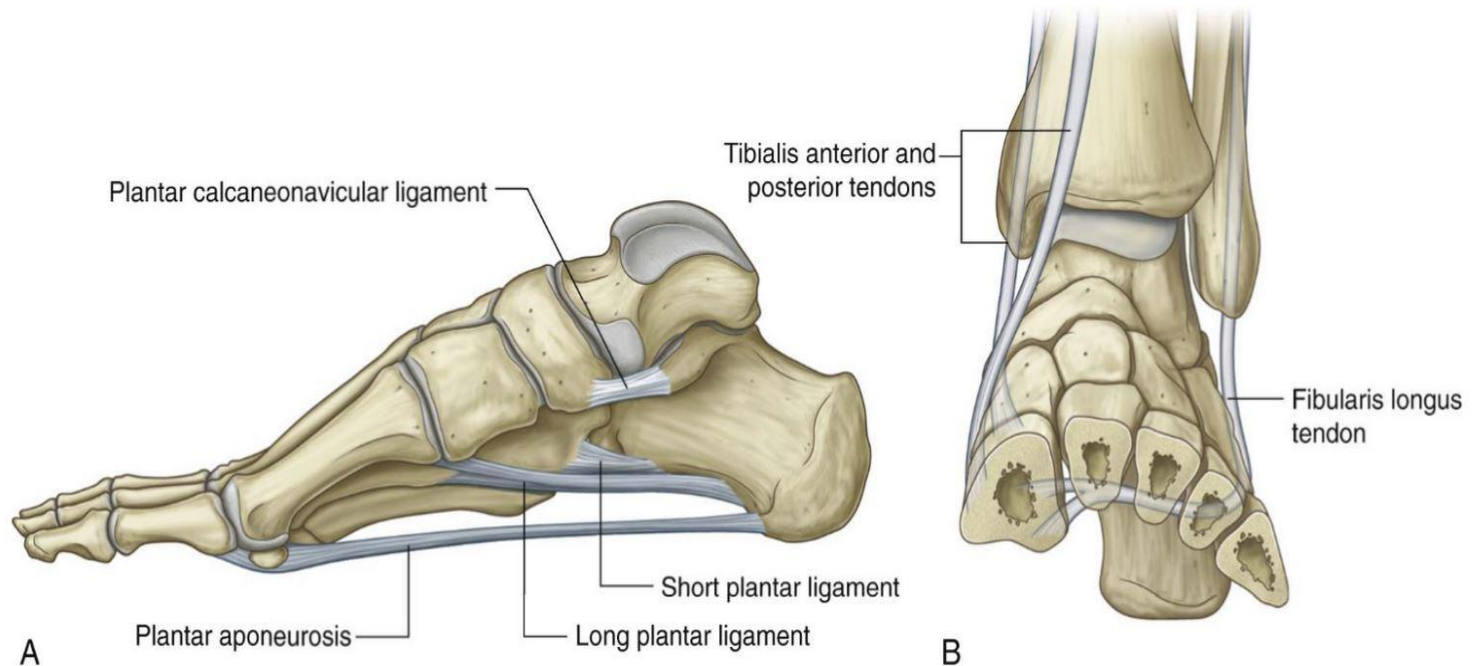


Arches of the Foot

- Bones of the foot do not lie in a horizontal plane. Instead, they form longitudinal and transverse arches relative to the ground
- Arches absorb and distribute downward forces from the body during standing and moving on different surfaces.



Arches of the Foot



Ligaments support the arches:

- plantar calcaneonavicular (spring ligament)
- plantar calcaneocuboid (short plantar ligament)
- long plantar ligaments
- plantar aponeurosis

Muscles that provide dynamic support to the arches:

- tendon of tibialis anterior
- tendon of tibialis posterior
- tendon of fibularis longus

Hallux Varus and Valgus

- Hallux Valgus- there is lateral deviation of the great toe in the transverse plane. It can be caused by pressure from the footwear and degenerative joint disease.



https://mdmercy.com/centers-of-excellence/orthopedics-bone-and-joint/institute-for-foot-and-ankle-reconstruction/conditions-we-treat/big-toe/hallux-valgus-bunion-deformity?sc_lang=en

- Hallux Varus - there is medial deviation of the great toe in the transverse plane. It can be caused by excessive recession of a bunion.



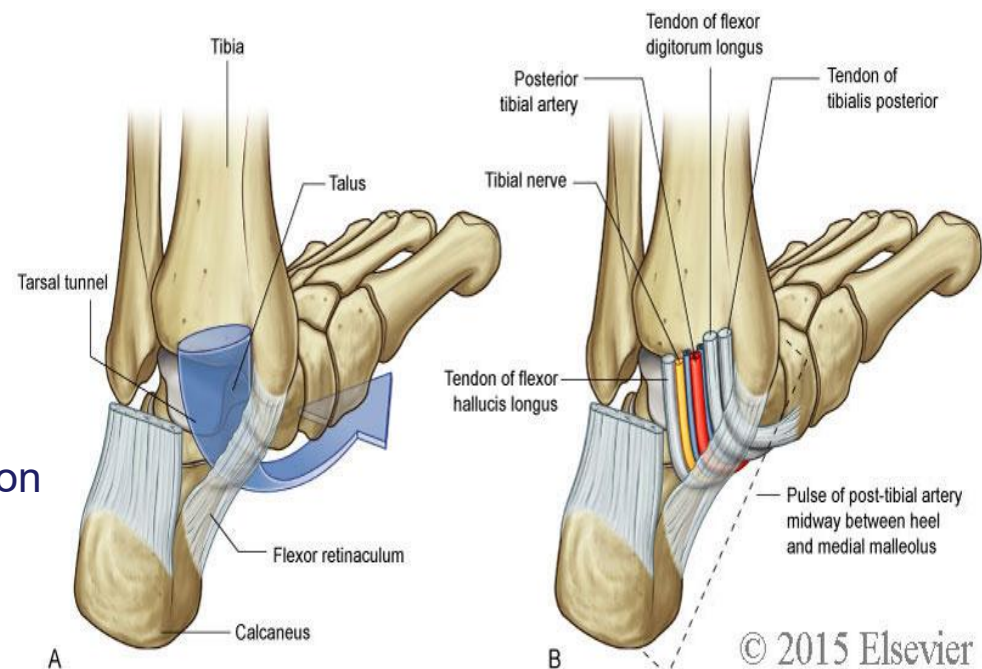
<https://mdmercy.com/centers-of-excellence/orthopedics-bone-and-joint/institute-for-foot-and-ankle-reconstruction/conditions-we-treat/big-toe/hallux-varus>

Medial Tarsal Tunnel – the gateway to the plantar surface of the foot

- Found on the posteromedial side of the ankle. Structures from the posterior leg enter the plantar foot.
- Formed by medial malleolus of the tibia, medial and posterior surfaces of the talus, medial surface of the calcaneus, inferior surface of the sustentaculum of the calcaneus.
- Covered by flexor retinaculum.

Contents of tarsal tunnel:

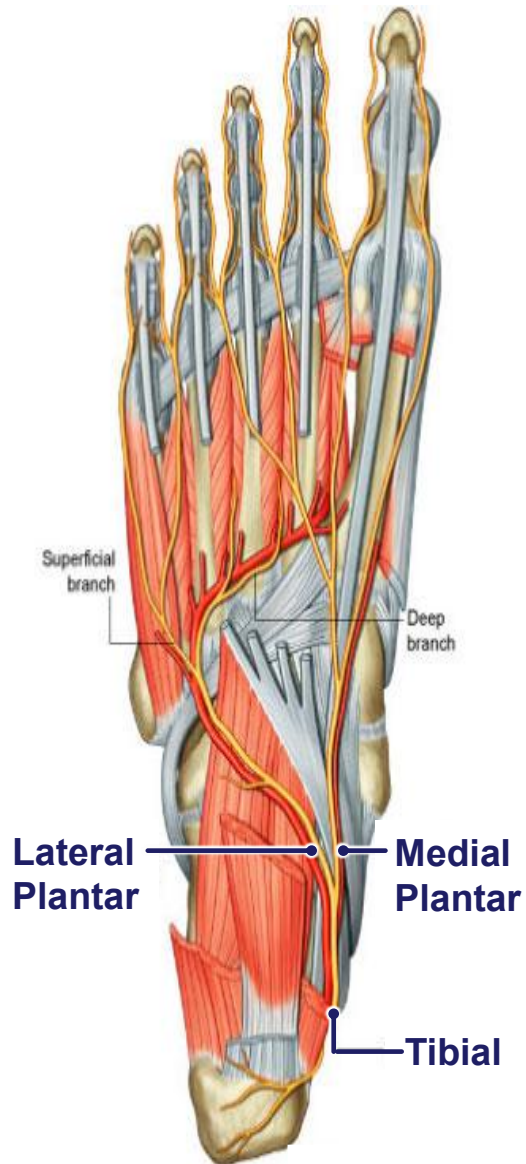
Tom **tibialis** posterior tendon
Dick flexor **digitorum** longus tendon
And posterior tibial **artery**
Very posterior tibial **vein**
Very posterior tibial **vein**
Nervous tibial **nerve**
Harry flexor **hallucis** longus tendon



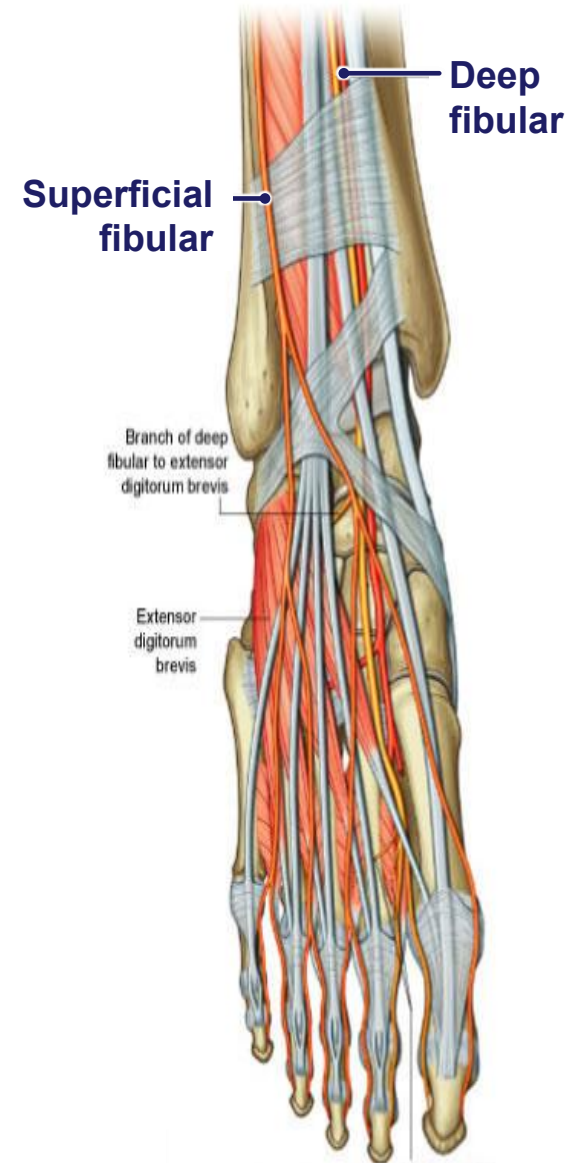
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Nerves of the Foot

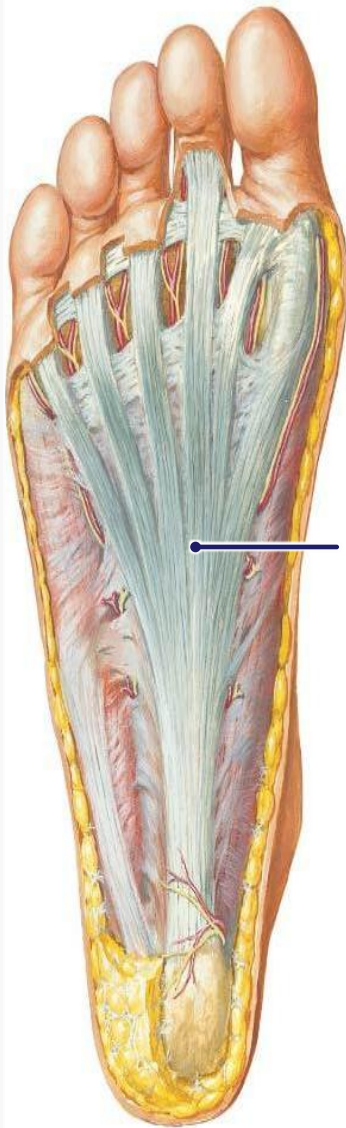


Plantar Surface



Dorsal Surface

Muscles of Plantar Foot - Layer 1



Plantar aponeurosis

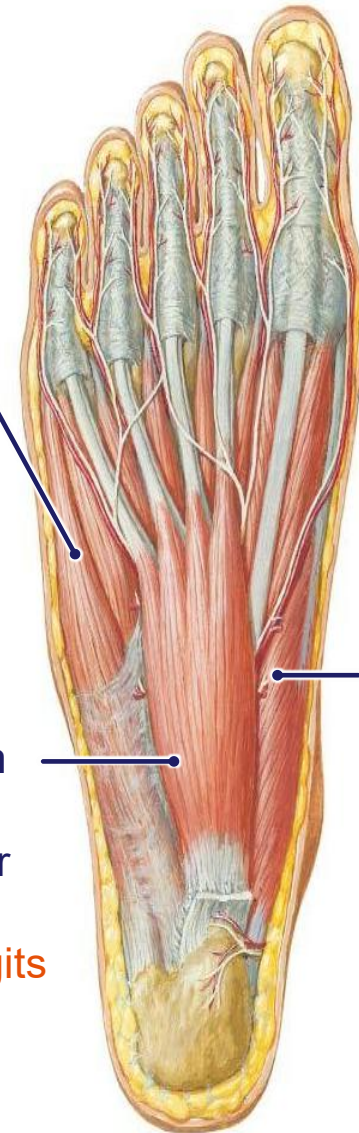
- supports the longitudinal arch of the foot
- protects deeper structures in the sole

Abductor digiti minimi

- lateral plantar nerve
- **abducts** the little toe

Flexor digitorum brevis

- medial plantar nerve
- **flexes** the digits



Abductor hallucis

- medial plantar nerve
- **abducts** the big toe

Muscles of Plantar Foot - Layer 2

Lumbricals

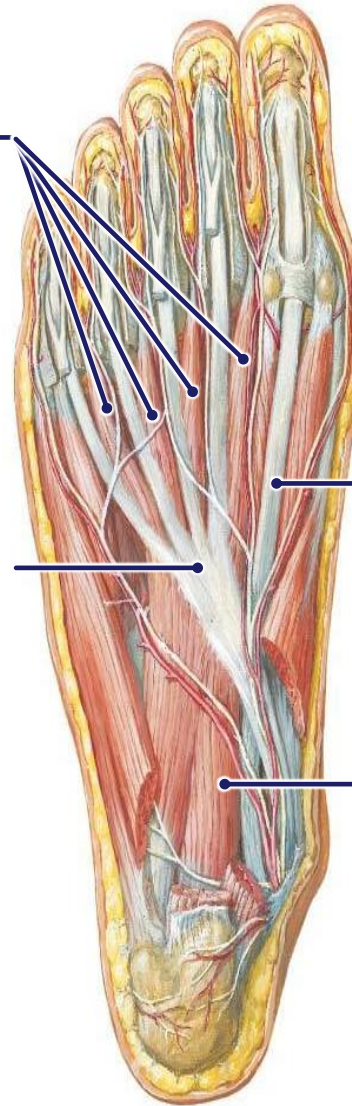
- 2nd-4th : lateral plantar
- 1st : medial plantar
- flexion at the MTP and extension at IP

Tendon of flexor digitorum longus

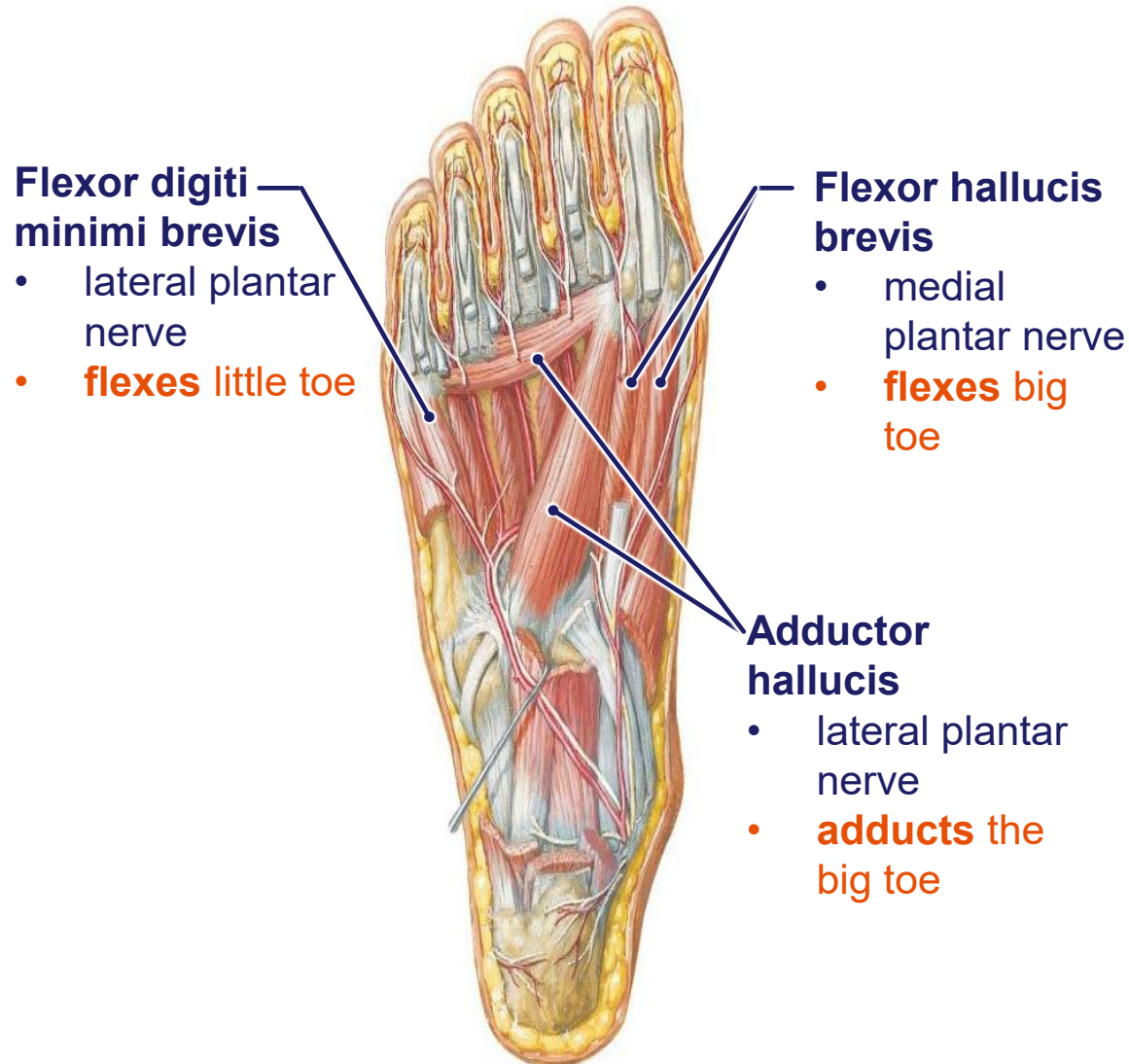
Tendon of flexor hallucis longus

Quadratus plantae

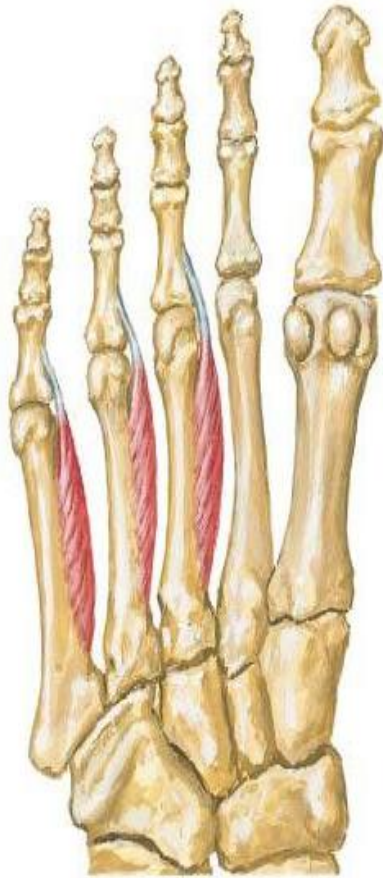
- lateral plantar nerve
- assist in flexion of digits



Muscles of Plantar Foot - Layer 3



Muscles of Plantar Foot - Layer 4



Plantar interossei - 3 Pad

- lateral plantar
- **aDductors**
- flexion at the MTP and extension at IP



Dorsal interossei - 4 Dab

- lateral plantar
- **aBductors**
- flexion at the MTP and extension at IP

Summary of Muscles of the Plantar Foot

Layer I

abductor hallucis
(A)

flexor digitorum
brevis (F)

abductor digiti
minimi



Layer II

quadratus
plantae

1st lumbrical (L)
2nd lumbrical
3rd lumbrical
4th lumbrical



Layer III

flexor hallucis
brevis (F)

adductor hallucis

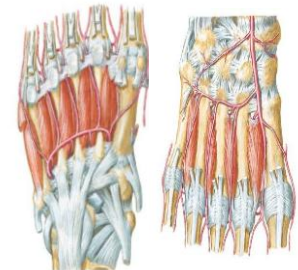
flexor digiti
minimi brevis



Layer IV

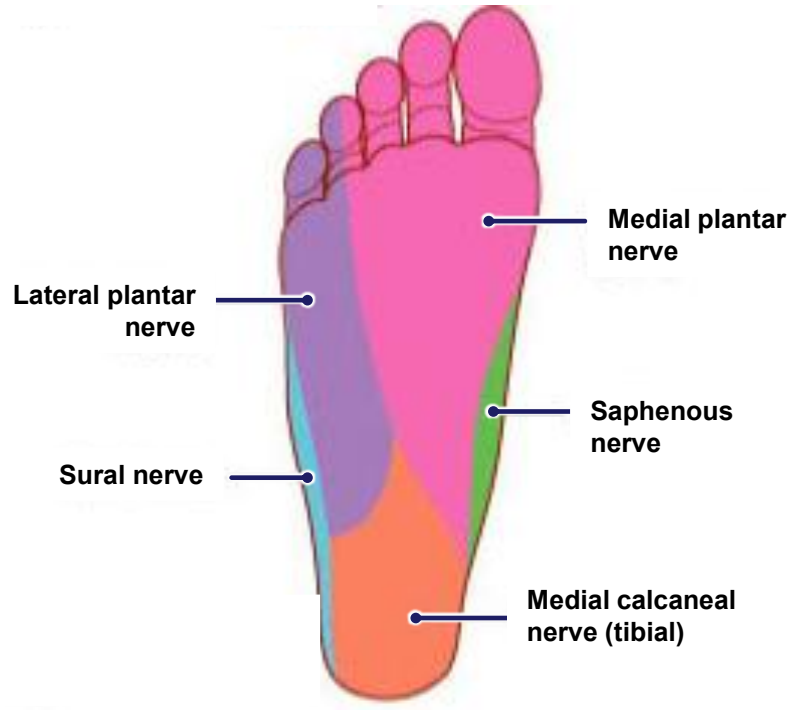
dorsal interossei

plantar
interossei

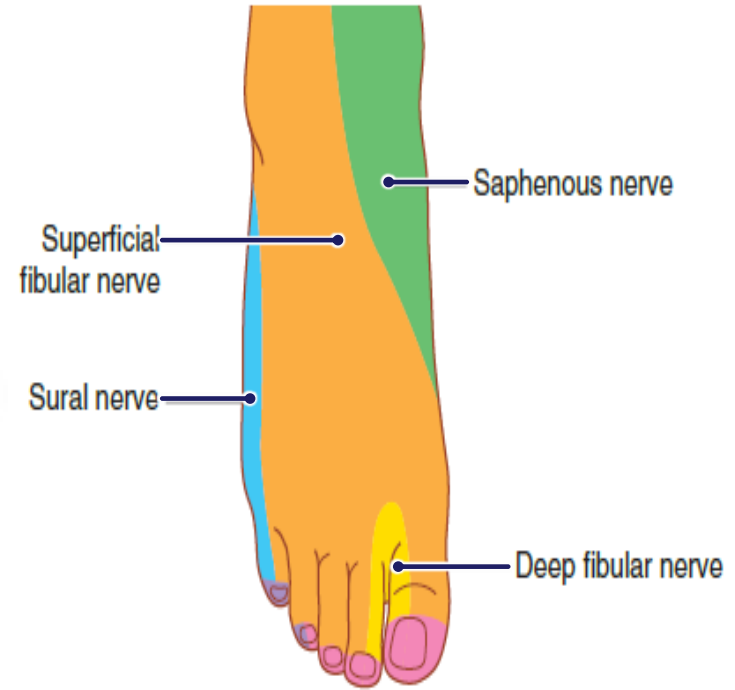


LAFF muscles – innervated by medial plantar, all other muscles
innervated by lateral plantar

Cutaneous Innervation of the Foot

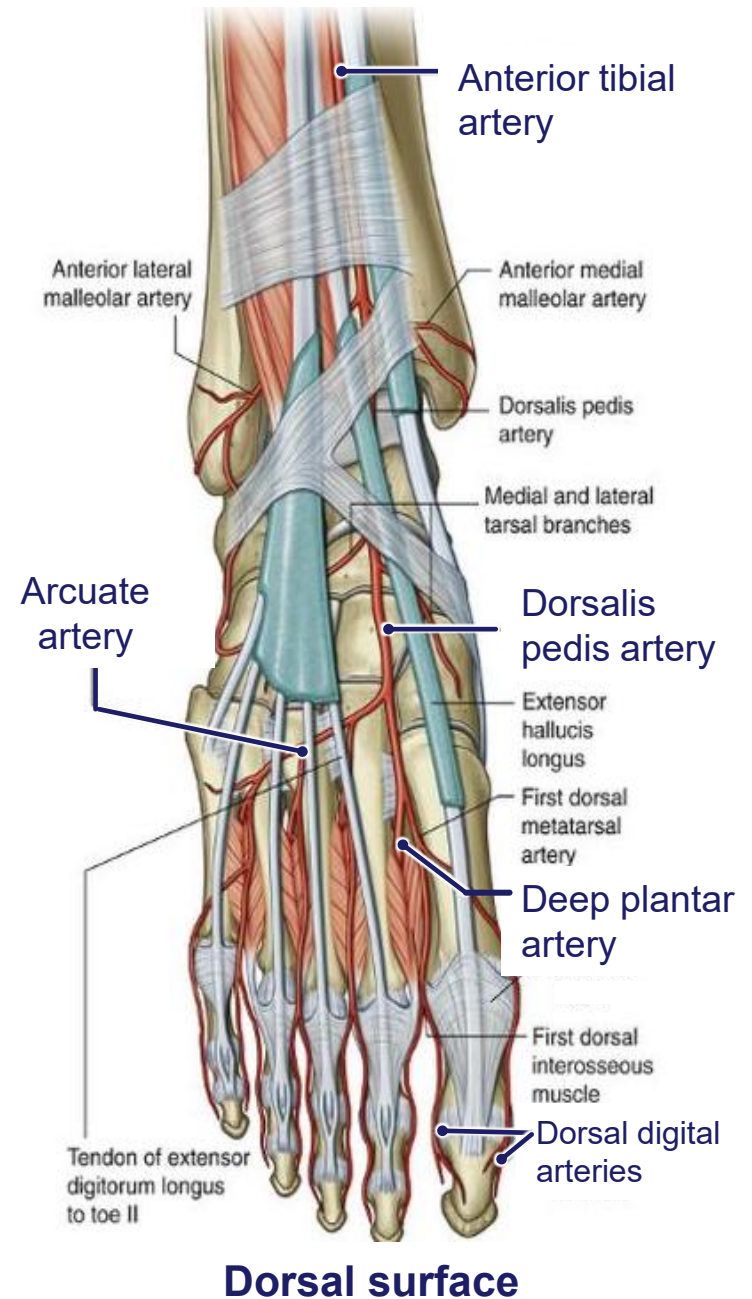
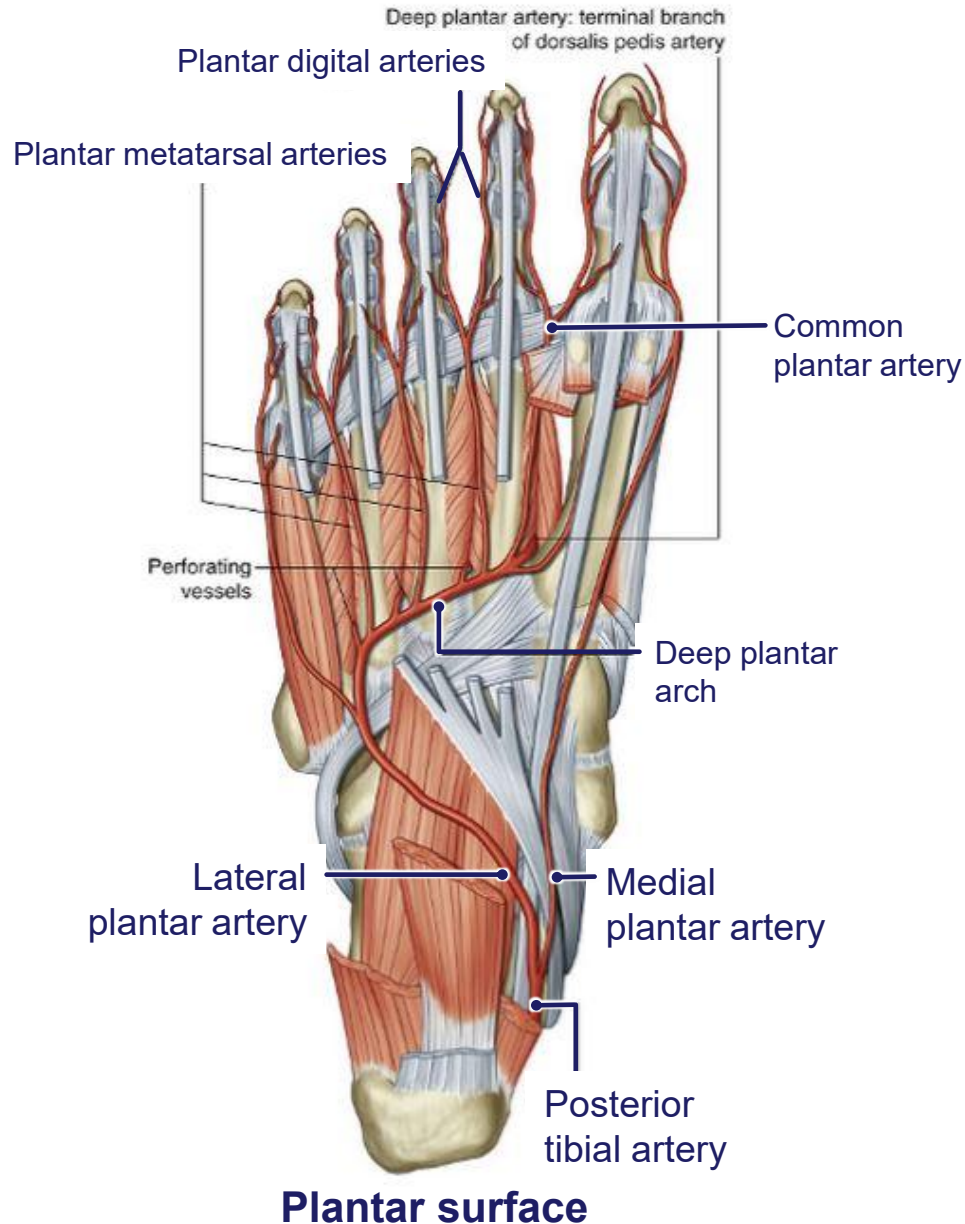


Plantar surface

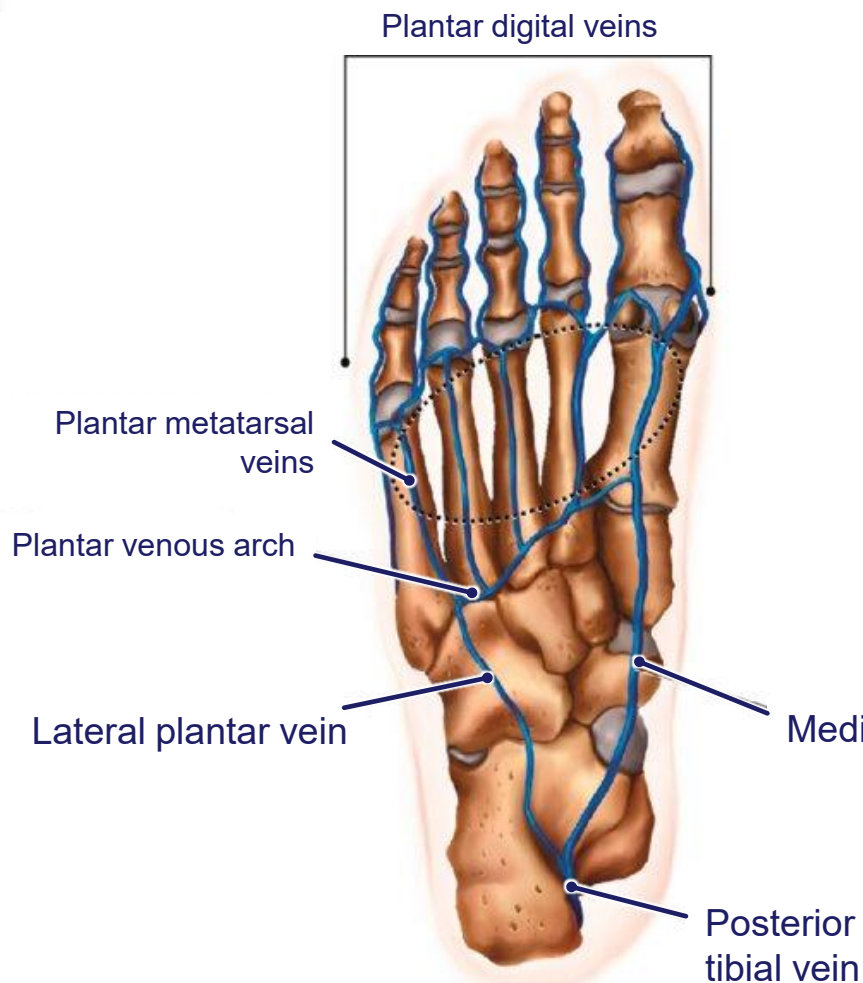


Dorsal surface

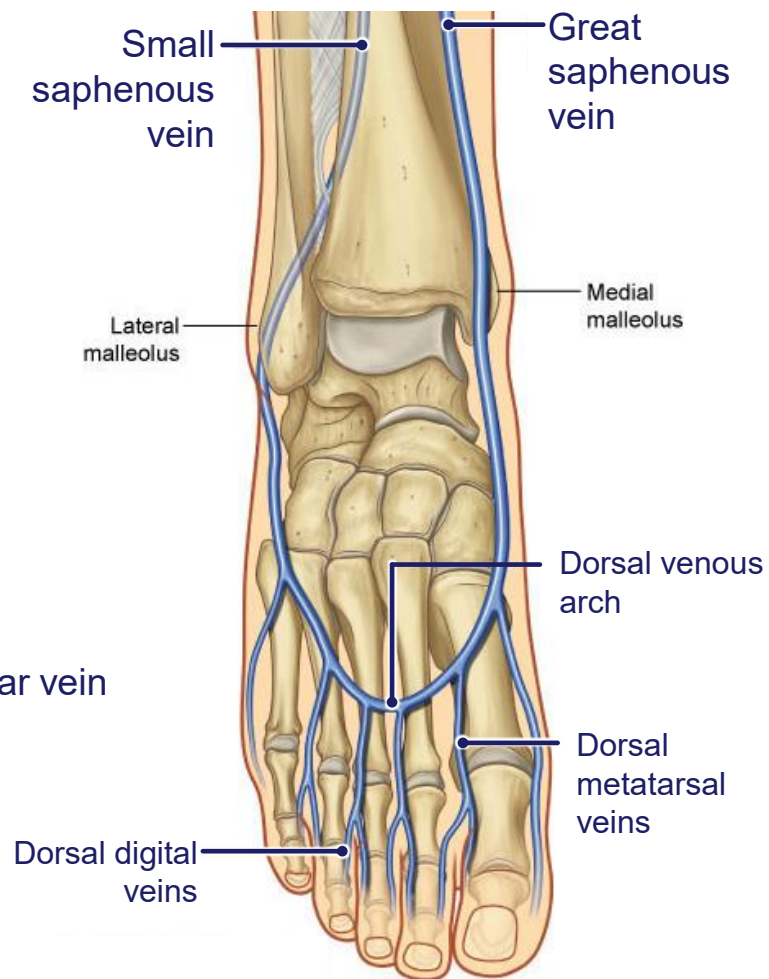
Arteries of the Foot



Veins of the Foot



Plantar surface

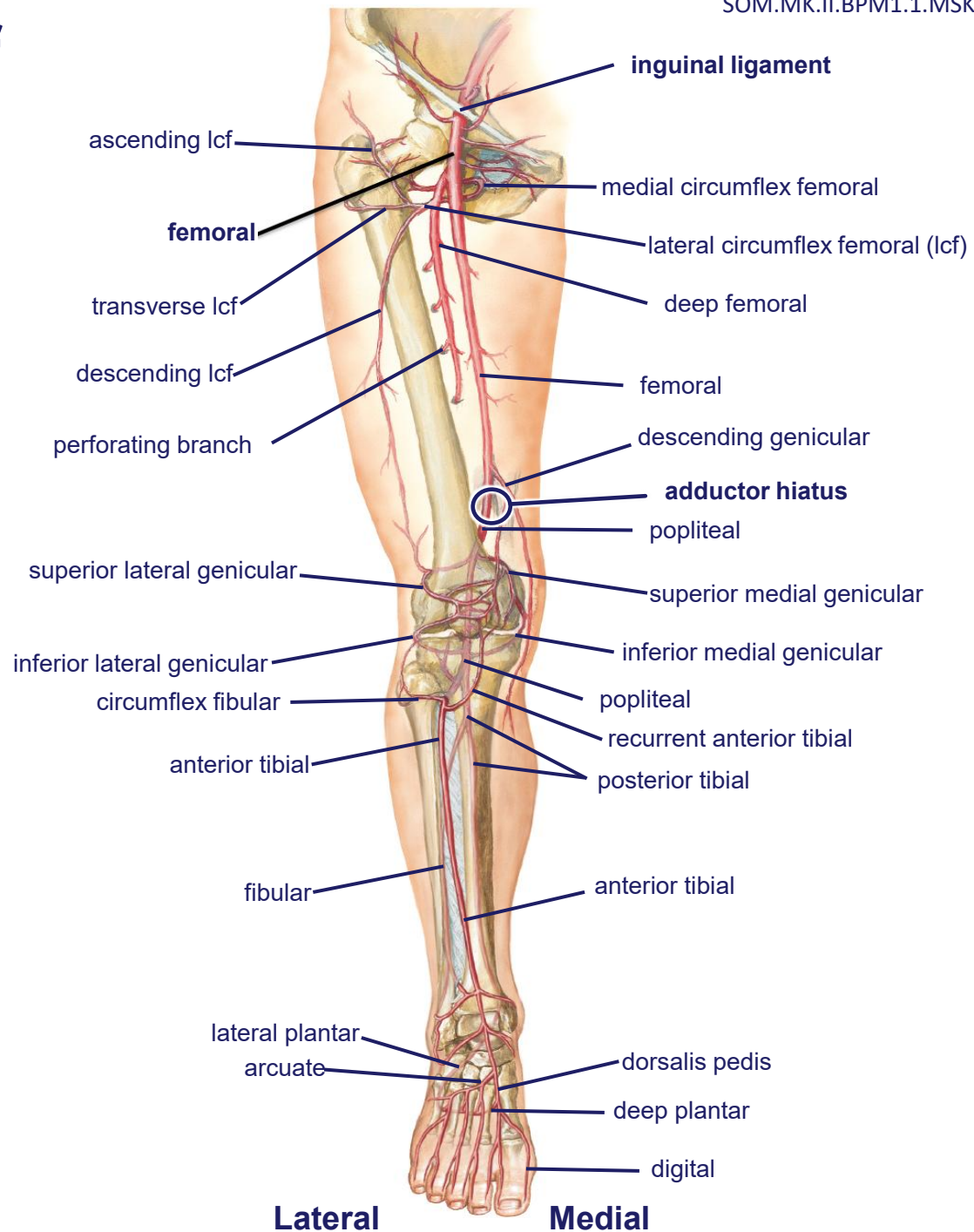


Dorsal surface

Summary of Arterial Supply of the Lower Limb

Summary of arterial supply of lower limb

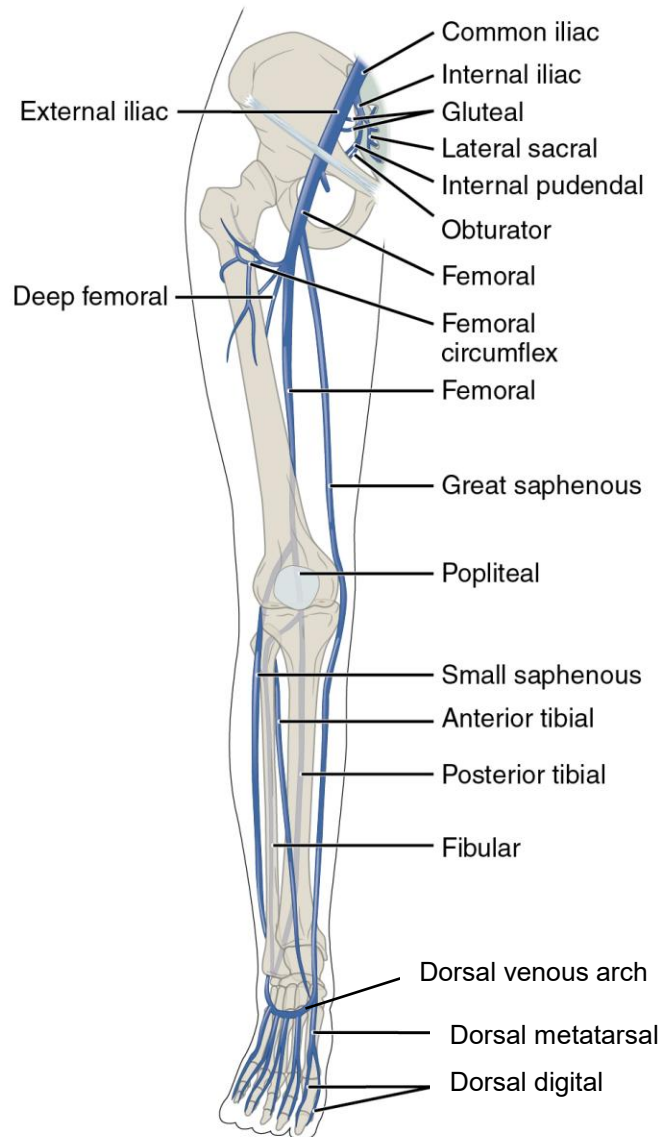
Now that we have completed the lower limb, use this slide to summarize the blood supply to the lower limb



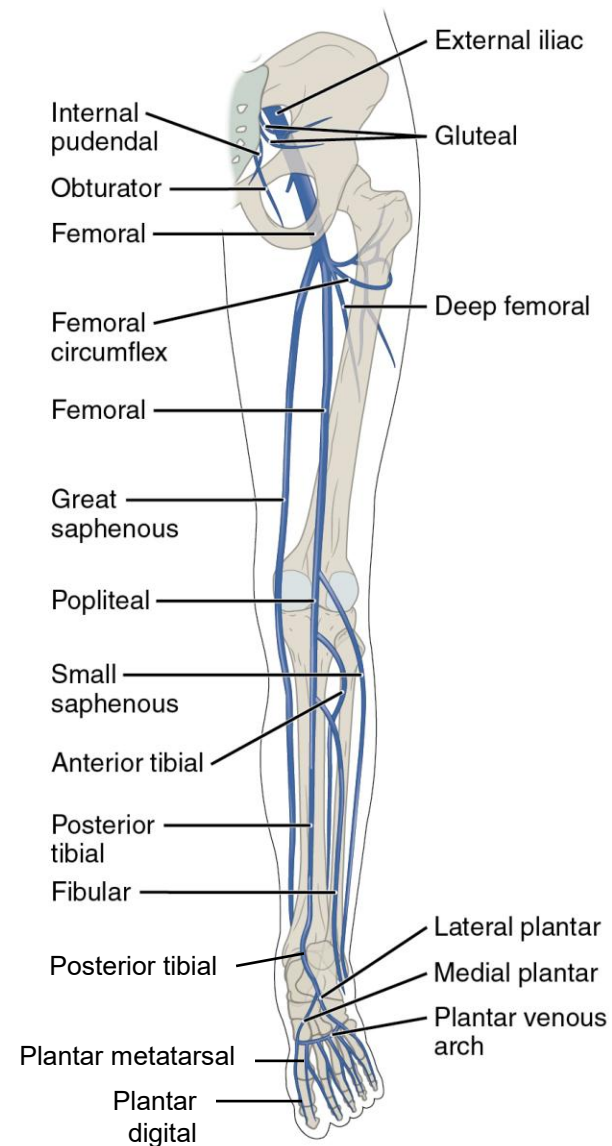
Summary of Venous Drainage of the Lower Limb

Summary of venous drainage of lower limb

Now that we have completed the lower limb, use this slide to summarize the venous drainage to the lower limb



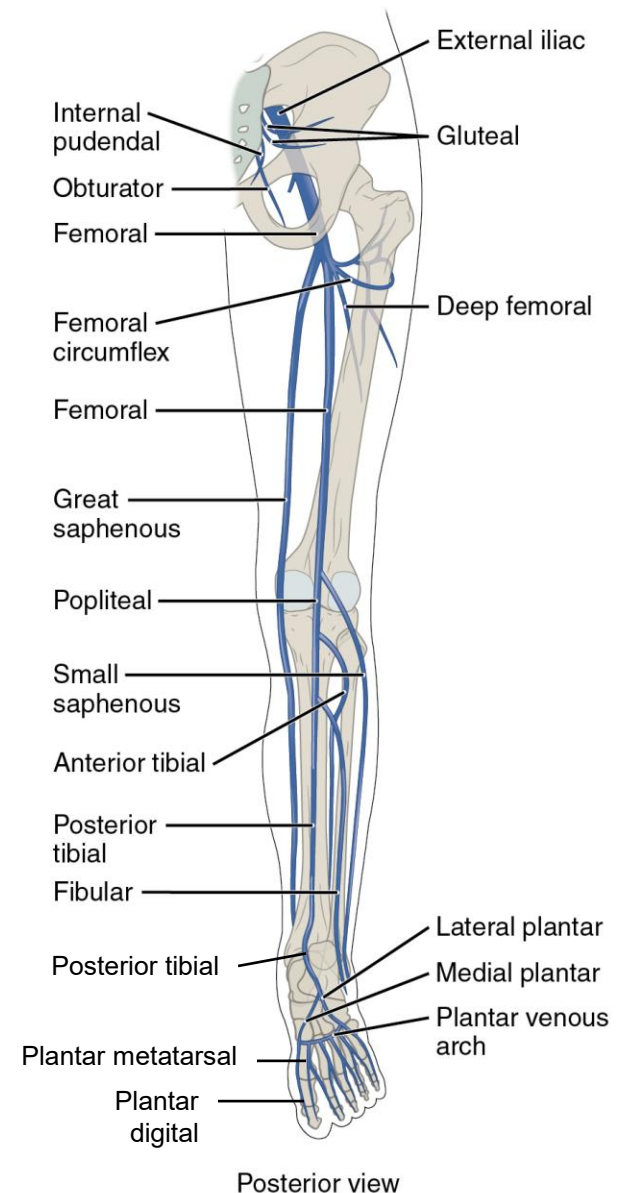
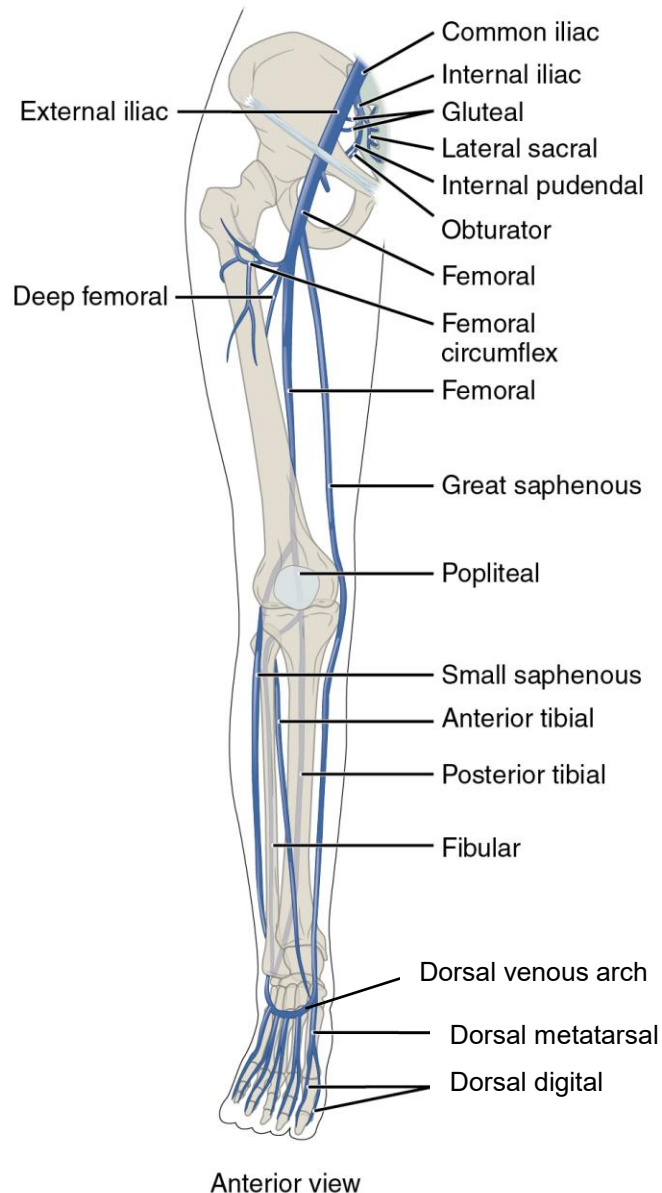
Anterior view



Posterior view

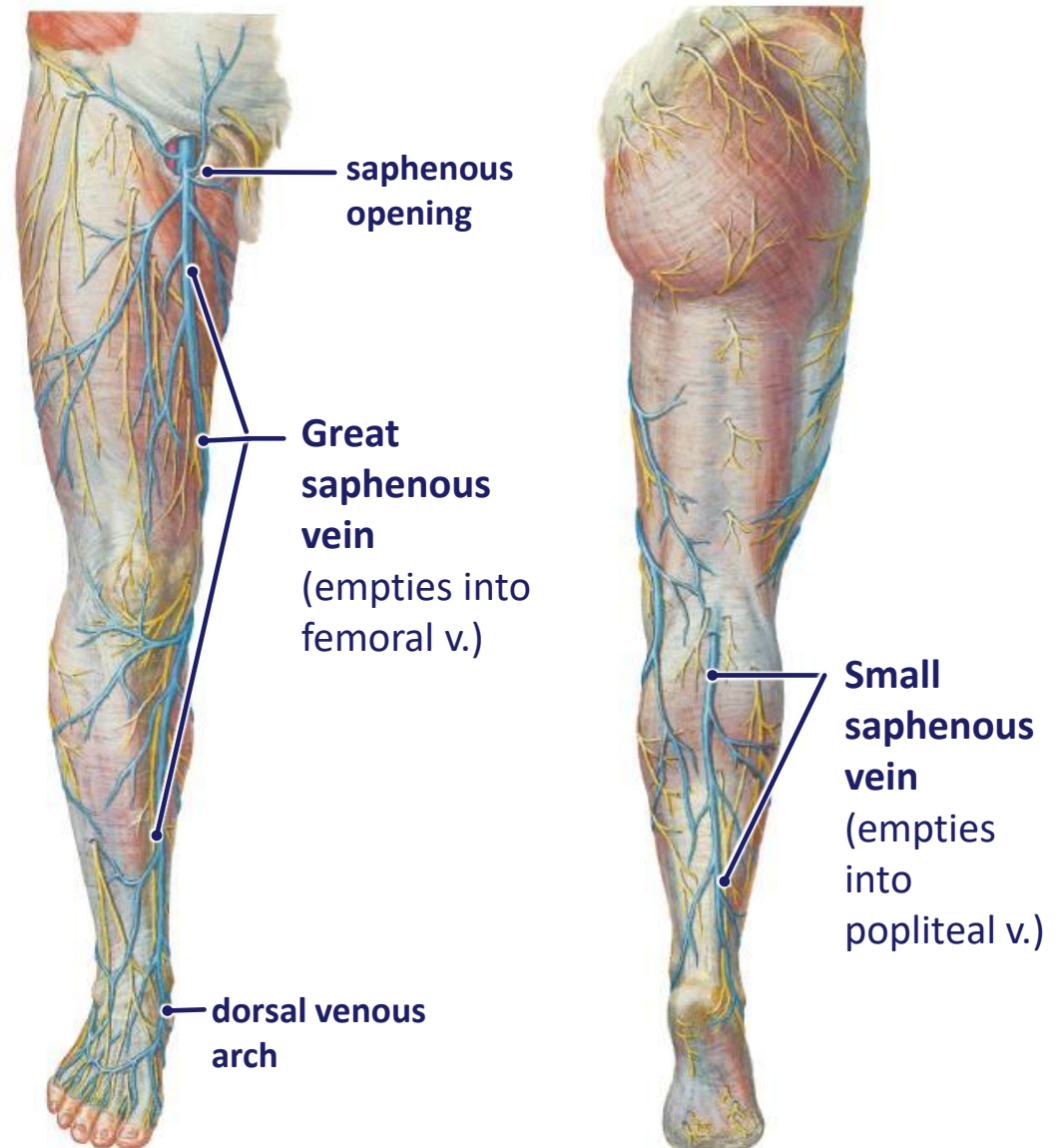
Deep Veins of the Lower Limb

- Located in the compartments of the lower limb
- Deep to the deep fascia (located in muscle compartments) – pressure rises when muscles contract = “muscular pump”
- **Runs with the corresponding arteries - same name and usually paired**
- Contains valves – prevents retrograde flow
- Receives blood from the superficial veins via “perforating” veins



Superficial Veins of the Lower Limb

- Located in the skin and superficial fascia of the lower limb
- Usually visible to the naked eye
- Outside deep fascia – not exposed to compression forces when muscles contract
- Contains valves –
 - prevents retrograde flow
 - prevents flow from deep veins to superficial veins
- Drain into deep veins
- *Venous cutdown in great saphenous vein: what nerve can be damaged due to this procedure?



Summary of Lymphatic Drainage of the Lower Limb

Lymphatic Drainage of the Lower Limb

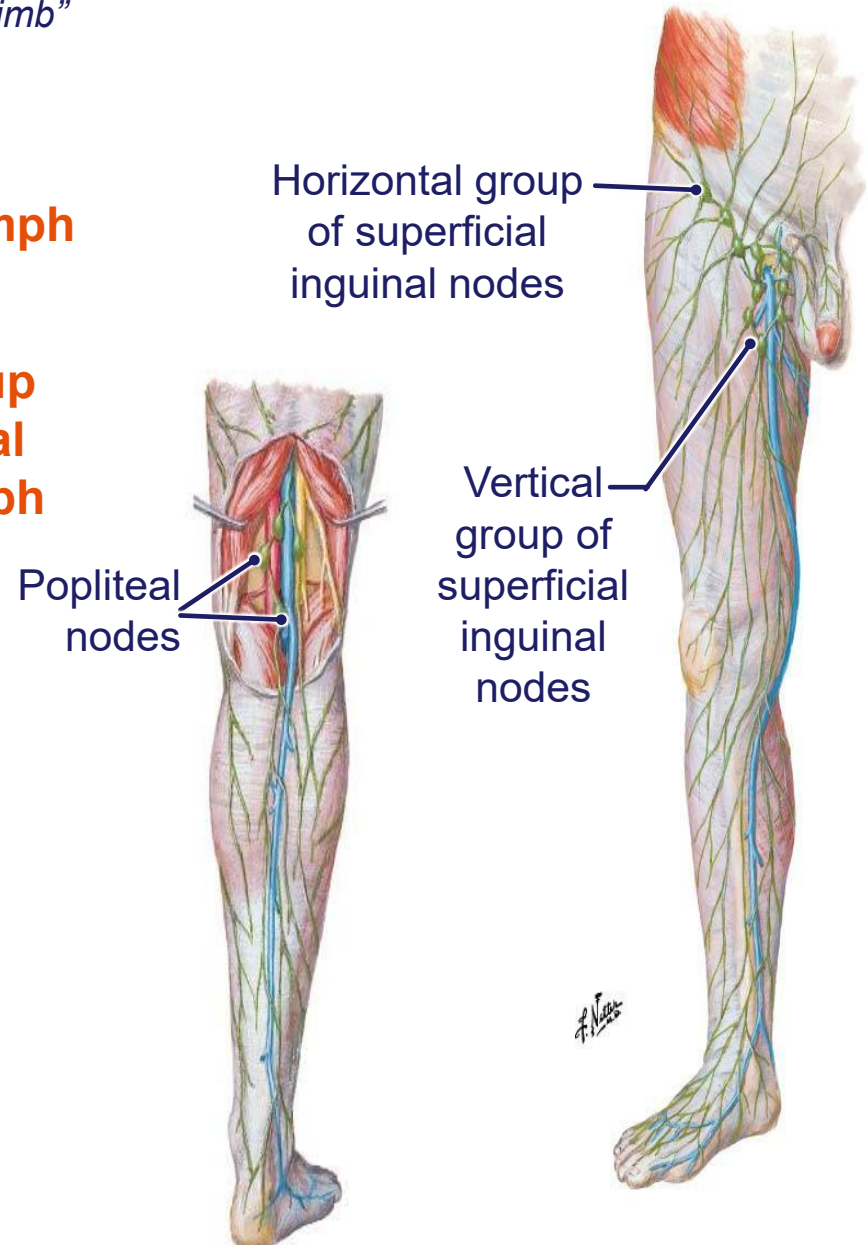
"Follows the path of the venous drainage of the limb"

Superficial lymphatic drainage:

- lateral foot
 - postero-lateral leg
- Popliteal lymph nodes**
- antero-medial foot
 - antero-medial leg
 - thigh
- Vertical group of superficial inguinal lymph nodes**

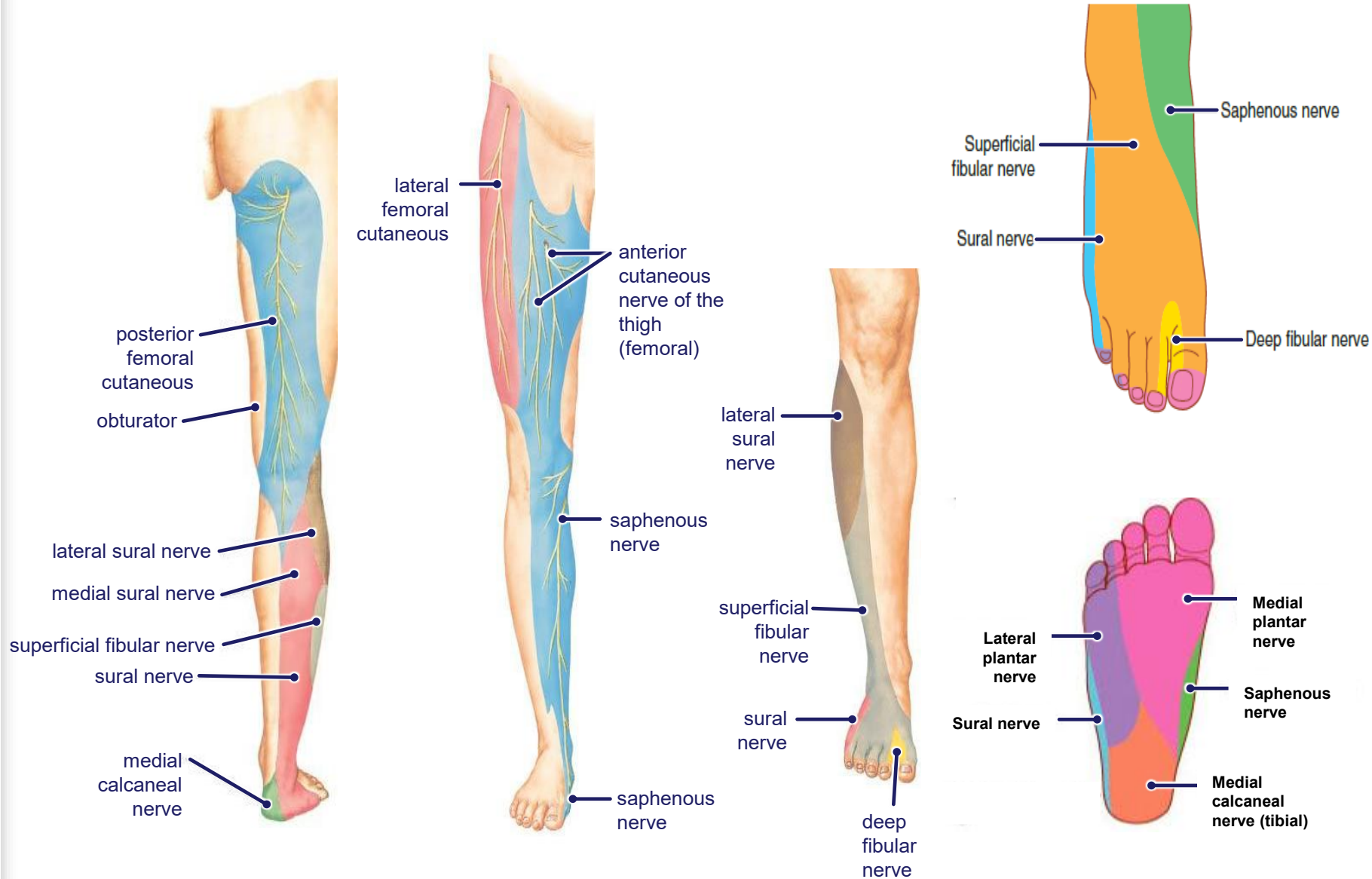
Deep lymphatic drainage

- leg and foot to popliteal nodes, then to deep inguinal
- thigh to deep inguinal nodes
- receives tributaries from the superficial groups and eventually drain into the external iliac nodes



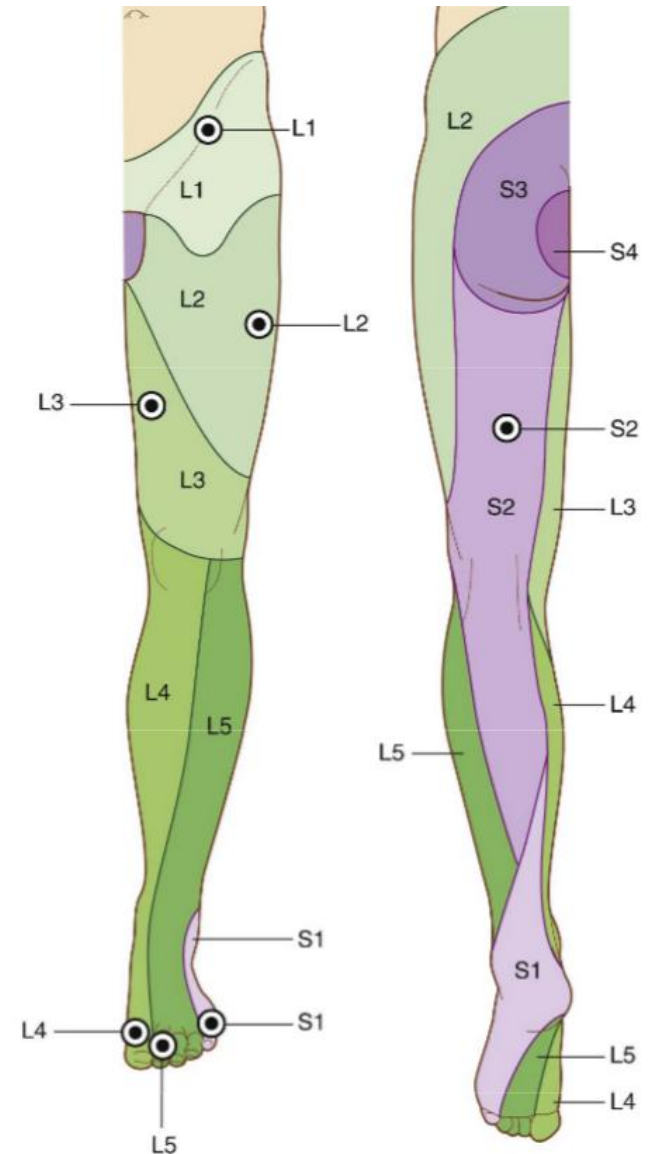
Summary of Cutaneous Innervation of the Lower Limb

Cutaneous Nerves of the Lower Limb



Dermatomes of Lower Limb

- Dermatomes in the lower limb and regions that can be tested for sensation and are reasonably autonomous (have minimal overlap) are:
 - over the inguinal ligament - L1,
 - lateral side of the thigh - L2,
 - lower medial side of the thigh - L3,
 - medial side of the great toe (digit I) - L4,
 - medial side of digit II - L5,
 - little toe (digit V) - S1,
 - back of the thigh - S2, and
 - skin over the gluteal fold - S3.



Myotomes of Lower Limb

Movements Generated by Myotomes

Selected myotomes of lower limb

- Flexion of the hip is controlled primarily by L1 and L2.
- Extension of the knee is controlled mainly by L3 and L4.
- Knee flexion is controlled mainly by L5 to S2.
- Plantarflexion of the foot is controlled predominantly by S1 and S2.
- Dorsiflexion of the foot is controlled predominately by L4 and L5.
- Adduction of the digits is controlled by S2 and S3.

